Multi-Modal Real-Time Driver Drowsiness and Fatigue Detection via 3D Facial Mesh Geometry, Head Pose Dynamics, and Hybrid Convolutional Neural Network Consensus

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***Abstract*—**Driver drowsiness, cognitive exhaustion, and involuntary microsleep episodes are among the most catastrophic contributing factors to vehicular crashes worldwide. Conventional vision-based driver monitoring frameworks heavily depend on classical Haar feature cascades or standalone deep convolutional neural networks (CNNs), both of which suffer from severe operational vulnerabilities: Haar cascades experience total detection collapse when eyes are closed because dark pupil contrast patterns vanish, while standalone CNN pipelines introduce high per-frame inference latency and are susceptible to input normalization defects. In this research, we present an end-to-end, multi-modal, real-time driver fatigue monitoring framework that unifies 478 dense 3D facial landmarks from MediaPipe FaceLandmarker with geometric temporal tracking, Perspective-n-Point (solvePnP) head pose dynamics, and hybrid CNN consensus verification. The proposed system continuously evaluates scale-invariant Eye Aspect Ratio (EAR) across 12 ocular landmark vectors, Mouth Aspect Ratio (MAR) across 4 oral aperture landmarks for persistent yawn detection, and computes real-time head pitch and roll to detect microsleep head drooping. Furthermore, we uncover and resolve a critical double-normalization bug prevalent in legacy architectures that previously caused severe CNN signal corruption. An asynchronous, non-blocking alert state machine incorporating a 0.3-second temporal persistence tolerance buffer is designed to eliminate false resets and guarantee alert actuation within 1.5 seconds of sustained eye closure. On standardized benchmark evaluations and live video tests, the proposed architecture achieves a 96.56% eye closure detection accuracy, 94.20% yawn detection accuracy, an ROC-AUC of 0.984, and a sustainable throughput of 32.6 frames per second (FPS) on standard multi-core CPUs without requiring dedicated GPU acceleration. Dual operational deployments—a native high-FPS OpenCV desktop HUD and a dynamic Streamlit telemetry dashboard—are demonstrated, establishing a robust, commercially viable active vehicle safety solution.

***Keywords*—**Driver Drowsiness Detection, Eye Aspect Ratio (EAR), Mouth Aspect Ratio (MAR), Head Pose Estimation, MediaPipe FaceMesh, Convolutional Neural Networks (CNN), Active Vehicle Safety, Microsleep Prevention.

I. INTRODUCTION

A. Background and Global Societal Impact

Transportation safety remains one of the paramount socio-economic challenges of modern civilization. Empirical accident investigations conducted by the World Health Organization (WHO), the National Highway Traffic Safety Administration (NHTSA), and the European Commission consistently reveal that operator fatigue and cognitive drowsiness account for approximately 20% to 30% of all fatal and catastrophic commercial transit accidents globally [1], [2]. Unlike impaired driving resulting from chemical intoxication, drowsiness exhibits an insidious onset: drivers frequently misjudge their subjective state of vigilance, falling victim to involuntary microsleep episodes.

A *microsleep* is clinically defined as an unintended, transient loss of consciousness lasting anywhere from 1.0 to 3.0 seconds, characterized by complete sensory deprivation and cognitive detachment. For a motor vehicle navigating at a conventional highway velocity of 100 km/h (27.8 m/s), a single 2.0-second microsleep implies that the vehicle travels over 55 meters entirely devoid of human steering, braking, or collision-avoidance intervention. The severity of resulting collisions is magnified because the driver makes no preemptive braking maneuvers before impact. Consequently, developing automated, non-invasive, cost-effective Driver Monitoring Systems (DMS) capable of preemptively alerting operators prior to critical failure is an urgent imperative in modern intelligent transportation systems (ITS).

B. Taxonomy of Driver Monitoring Approaches

Contemporary driver fatigue monitoring paradigms are categorized into three major technical classifications:

1) *Vehicle Telematics and Driving Dynamics:* Systems monitoring steering wheel angle variability, standard deviation of lane position (SDLP), and pedal actuation dynamics. While non-intrusive, these systems exhibit high detection latency: by the time steering jitter or lane drifting occurs, the driver has already lost physical control of the vehicle, leaving dangerously narrow reaction margins.

2) *Intrusive Physiological Sensor Networks:* Approaches utilizing electroencephalography (EEG), electrocardiography (ECG), electrooculography (EOG), or galvanic skin response (GSR). Although physiological signals yield near-perfect diagnostic accuracy in clinical laboratories, their physical implementation requires adhesive electrodes, headgear, or wristbands that cause skin irritation and operational impedance, rendering them unacceptable to commercial drivers.

3) *Computer Vision and Facial Geometry:* Non-intrusive optical systems capturing facial features through interior cabin cameras. Optical systems analyze ocular dynamics (blink duration, closure frequency), oral dynamics (yawning aperture), and postural dynamics (head tilting and nodding). Computer vision represents the optimal compromise between driver comfort, early fatigue intervention, and non-contact operational reliability.

C. Critical Vulnerabilities in Existing Vision Pipelines

Despite decades of computer vision research, standard academic and open-source driver drowsiness systems suffer from fundamental structural vulnerabilities:

1) The Eye-Closure Tracking Collapse of Haar Cascades: A widespread paradigm in student and baseline academic literature relies on OpenCV's Viola-Jones Haar feature-based cascades (e.g., `haarcascade_eye_tree_eyeglasses.xml`). However, Haar eye cascades are fundamentally trained to identify open eyes based on high-contrast circular edges formed by the dark pupil/iris contrasted against the white sclera. **When a driver closes their eyes, Haar cascades return zero detections**. In naive code implementations, downstream neural network prediction and tracking loops are conditioned on positive eye bounding boxes (i.e., `if len(eyes) > 0:`). As an alarming direct consequence, *the moment the driver closes their eyes to sleep, the software stops tracking eyes entirely*, failing to ever register an "Eyes Closed" state.

2) Destructive Normalization Bugs: As uncovered during our codebase audit, deep learning models exported from Keras/TensorFlow frequently incorporate built-in preprocessing layers, such as `Rescaling(1/255.0)`. When legacy application wrappers manually divide raw pixel values by 255.0 prior to inference, the model divides by 255 a second time ($x / 65025.0$). This arithmetic error suppresses feature activations to near zero, degrading model sensitivity into random guessing.

3) Rigid Countdown Timer Resets: Baseline alert systems commonly utilize naive time accumulators without hysteresis or persistence filtering. A single dropped video frame, momentary tracker jitter, or minor head motion resets the elapsed closure timer back to zero, preventing alarms from ever sounding during genuine microsleep events.

D. Primary Contributions of This Study

To address the aforementioned systemic limitations, this paper proposes an intelligent, multi-modal driver drowsiness detection system with the following validated contributions:

- **Dense 3D Geometric Mesh Tracking:** We deploy MediaPipe FaceLandmarker with 478 3D landmarks operating on vectorized CPU delegates, ensuring that eye and lip contours are continuously tracked even under total eyelid closure and varying head orientations.
- **Multi-Modal Geometric Fusion:** We formulate a unified monitoring model that fuses Eye Aspect Ratio (EAR), Mouth Aspect Ratio (MAR), and Perspective-n-Point (solvePnP) head pitch estimation to capture the complete spectrum of fatigue manifestations: microsleep, yawning, and head drooping.
- **Algorithmic Rectification of Legacy Defects:** We systematically diagnose and resolve the double-normalization arithmetic bug, implementing dynamic introspection to ensure exact scaling across heterogeneous CNN backends.
- **Asynchronous Robust Alert State Machine:** We engineer a multi-threaded alert engine with a 0.3-second temporal persistence tolerance buffer, eliminating false resets and guaranteeing alarm actuation at 1.5s of sustained eye closure.
- **Dual-Interface Operational Deployment:** We implement both a high-throughput C++-accelerated OpenCV desktop interface (30–60 FPS) with a full visual HUD and a reactive Streamlit web dashboard for remote telemetry.

II. LITERATURE SURVEY

A. Foundational Vision Detectors

The pioneering contribution to real-time face detection was formulated by Viola and Jones [3] utilizing integral images, AdaBoost learning, and Haar-like rectangular filters. While computationally revolutionary for early embedded processors, Haar cascades exhibit high vulnerability to out-of-plane head rotations, ambient cabin shadows, and morphological eye differences. Most critically, because Haar cascades detect contrast gradients between the iris and sclera, they completely fail when the eye is closed, rendering them unsuited as standalone sleep detectors.

Kazemi and Sullivan [4] introduced an ensemble of regression trees to estimate facial landmark positions directly from pixel intensities in one millisecond. The Dlib 68-landmark model popularized this approach. However, Dlib's 68-point model is sensitive to partial occlusions and requires high CPU overhead compared to modern mobile-first graph networks.

B. Geometric Aspect Ratio Formulations

Soukupová and Čech [5] formulated the landmark-based Eye Aspect Ratio (EAR) metric, which parameterized eyelid aperture into a continuous scalar. Their seminal work proved that EAR is scale-invariant and maps eye opening directly to a scalar distance ratio, accurately separating brief involuntary blinks (100–400 ms) from prolonged microsleeps. Tereza et al. [6] conducted extensive highway validation, establishing that normal spontaneous blinking spans 100 to 400 milliseconds, whereas microsleep transitions occur when closure persists beyond 1.5 seconds.

For oral fatigue dynamics, Abtahi et al. [7] published the YawDD dataset and demonstrated that yawning frequency serves as a reliable precursor to cognitive degradation, often

occurring 10 to 15 minutes prior to severe ocular drowsiness. Bakheet and Al-Hamadi [8] proposed Gabor wavelets and SVMs for mouth state classification, though their reliance on manual thresholding introduced sensitivity to speech articulation. Modern landmark-based Mouth Aspect Ratio (MAR) approaches overcome this by computing relative vertical aperture over horizontal lip width.

C. Deep Learning and Ensembles in Driver Monitoring

With the rise of deep convolutional neural networks (CNNs), researchers explored end-to-end classification of cropped eye regions. Reddy et al. [9] proposed a compressed MobileNet architecture for embedded vehicle boards. While reaching 92.5% accuracy, raw CNN classification evaluated on every video frame demands significant CPU cycles, causing frame rate throttling on commodity hardware. Hybrid ensembles combining fast geometric filters with deep neural networks have emerged as the optimal balance for real-time safety systems.

Table I presents a detailed literature survey matrix summarizing 15 foundational and state-of-the-art works, detailing their problems, methods, datasets, findings, and direct relevance to our system.

TABLE I. COMPREHENSIVE SURVEY OF RELATED RESEARCH IN DRIVER MONITORING SYSTEMS

Ref.	Author(s) & Year	Problem Addressed	Methodology	Dataset	Key Findings & Strengths	Identified Limitation / Gap	Relevance
[3]	Viola & Jones (2004)	Rapid facial feature detection	Haar-like cascades, AdaBoost	Custom Face DB	Real-time face detection at 15 FPS on early CPUs	Fails completely when eyes close; sensitive to lighting	Baseline comparison
[4]	Kazemi & Sullivan (2014)	Millisecond facial landmark alignment	Ensemble of regression trees (ERT)	LFPW, HELEN	Fast 68-point 2D facial landmark localization	Lacks 3D depth; tracking fails under steep head yaw/pitch	Landmark benchmark
[5]	Soukupová & Čech (2016)	Real-time eye blink & closure tracking	Facial landmark EAR formulation	Closed Eyes DB	Scale-invariant eye closure scalar; distinguishes blinks	Lacks oral and head pose fusion; single sensor failure	Core ocular formulation
[6]	Tereza et al. (2019)	Highway driver fatigue validation	EAR temporal windowing	Real Highway DB	Validated that blinks span 100-400ms while microsleep > 1.5s	Fixed thresholds fail across diverse ethnic ocular phenotypes	Timing thresholds
[7]	Abtahi et al. (2014)	Driver yawning and fatigue benchmarking	Geometric mouth aspect tracking	YawDD Dataset	Established benchmark dataset for driver yawning dynamics	High false positive rate during vocal conversation and singing	Oral yawn baseline
[8]	Bakheet & Al-Hamadi (2021)	Vision-based driver yawning detection	Gabor wavelets, SVM classifier	YawDD, NTHU-DDD	91.3% yawn detection accuracy under controlled illumination	Feature extraction computationally heavy for multi-core CPUs	Mouth aspect ratio
[9]	Reddy et al. (2017)	Embedded edge drowsiness detection	MobileNet CNN compression	Custom Driving Video	92.5% accuracy on embedded Nvidia Jetson edge boards	Frame-by-frame inference drops FPS below real-time on standard CPU	Throughput benchmark
[10]	Deng & Zhou (2018)	Head pose and eye state fusion	Multi-task cascaded CNN	NTHU-DDD Dataset	Combining head pose reduces false blink alarms by 14%	Complex multi-branch training; memory intensive	solvePnP head pose
[11]	Lugaresi et al. (2019)	Real-time on-device streaming perception	MediaPipe Face Mesh (468/478 pts)	Google Internal DB	Sub-millisecond landmark tracking with XNNPACK on CPU	Raw coordinates lack high-level fatigue state logic	Primary perception engine
[12]	You et al. (2019)	Multi-feature fusion for driver fatigue	PERCLOS + Yawn + Head Tilt	Static Laboratory DB	Achieved 94.3% classification accuracy via multi-cue fusion	Evaluated only on offline static frames; lacks live latency profiling	Multi-cue architecture
[13]	Cray et al. (2021)	False alarm filtering in driver alert systems	Temporal sliding persistence windows	Synthetic Driving DB	Buffer delays prevent single-frame false siren triggers	Static buffer sizes cause alert lag during rapid microsleep	Tolerance buffer design
[14]	Smith et al. (2018)	Microsleep pathophysiology and crashes	Clinical EEG vs Video correlation	Sleep Laboratory DB	Proved 1.5s to 2.0s eye closure correlates with stage-1 sleep EEG	Clinical study; no real-time embedded vision implementation	Clinical justification
[15]	Lepetit et al. (2009)	Fast accurate Perspective-n-Point	EPnP $O(n)$ formulation	Synthetic Point Cloud	Fast non-iterative 3D pose estimation from 2D correspondences	Requires known 3D facial CAD model calibration	solvePnP derivation
[16]	Dietterich (1998)	Statistical comparison of ML algorithms	McNemar's test and 5x2 CV	UCI Benchmarks	Established rigorous statistical validation protocol for classifiers	Theoretical evaluation; not applied to real-time video streams	Statistical tests
[17]	Lones (2021)	Pitfalls in machine learning research	Methodological checklist	Literature Survey	Documents data leakage, double-scaling bugs, and validation bias	Survey guideline without domain-specific vision implementation	Bug audit basis

III. MATERIALS AND METHODS

A. Architectural Flowchart and Pipeline

The end-to-end architecture of our multi-modal driver drowsiness detection system is structured into a modular pipelined topology, as illustrated in Fig. 1 and formal algorithm specification in Algorithm 1. The framework executes in real time without inter-frame blocking:

Algorithm 1: Real-Time Multi-Modal Driver Drowsiness and Fatigue Detection

Input: Video stream frame $\mathbf{I}_t \in \mathbb{R}^{H \times W \times 3}$, thresholds τ_{EAR}, τ_{MAR}, θ_{nod}, Δt_{sleep}

Output: Annotated frame $\mathbf{I}_{\text{annot}}$, Alert level $\mathcal{A}_t \in \{\text{NORMAL}, \text{WARNING}, \text{CRITICAL}\}$

Invert frame horizontally: $\mathbf{I}_{\text{flip}} \leftarrow \text{flip}(\mathbf{I}_t, 1)$; Convert color $\mathbf{I}_{\text{RGB}} \leftarrow \mathcal{C}_{\text{BGR}\to\text{RGB}}(\mathbf{I}_{\text{flip}})$

Localize 3D mesh: $\mathcal{M} = \{\mathbf{P}_i\}_{i=1}^{478} \leftarrow \text{FaceLandmarker}(\mathbf{I}_{\text{RGB}})$

if $\mathcal{M}$ is empty **then** return "Face Not Detected", $\mathcal{A}_t \leftarrow$ NORMAL **end if**

Compute $\text{EAR}_{\text{left}}, \text{EAR}_{\text{right}}$ using Eq. (3); $\text{EAR}_{\text{avg}} \leftarrow \frac{1}{2}(\text{EAR}_{\text{left}} + \text{EAR}_{\text{right}})$

Compute MAR across landmarks $\{13, 14, 78, 308\}$ using Eq. (5)

Compute head pitch θ_{pitch} via $\text{solvePnP}(\mathbf{X}_{\text{3D}}, \mathbf{x}_{\text{2D}}, \mathbf{K})$ using Eq. (6)–(7)

if CNN verification enabled **then** $P_{\text{closed}} \leftarrow \text{CNN}(\text{crop}(\text{eyes}))$; Update hybrid vote **end if**

if $S_{\text{eye}} == \text{Closed}$ **then**

  if $t_{\text{start}} == \text{None}$ **then** $t_{\text{start}} \leftarrow \text{timestamp}()$ **end if**

  $\Delta t_{\text{closed}} \leftarrow \text{timestamp}() - t_{\text{start}}$

  if $\Delta t_{\text{closed}} \ge \Delta t_{\text{sleep}}$ **then** $\mathcal{A}_t \leftarrow \text{CRITICAL}$; $\text{StartAlarm}()$

  else if $\Delta t_{\text{closed}} \ge \frac{1}{2}\Delta t_{\text{sleep}}$ **then** $\mathcal{A}_t \leftarrow \text{WARNING}$ **end if**

else

  if $\text{timestamp}() - t_{\text{last_closed}} > \delta_{\text{tol}}$ **then** $t_{\text{start}} \leftarrow \text{None}$; $\text{StopAlarm}()$ **end if**

  if $\text{MAR} > \tau_{\text{MAR}} \lor \theta_{\text{pitch}} < \theta_{\text{nod}}$ **then** $\mathcal{A}_t \leftarrow \text{WARNING}$ **end if**

end if

Render HUD overlays $\mathbf{I}_{\text{annot}} \leftarrow \text{DrawHUD}(\mathbf{I}_{\text{flip}}, \mathcal{A}_t, \text{EAR}, \text{MAR}, \theta_{\text{pitch}})$

return $\mathbf{I}_{\text{annot}}, \mathcal{A}_t$

B. Mathematical Derivation of Scale-Invariant EAR

To demonstrate why EAR is scale-invariant, let an eye be subject to an arbitrary isotropic camera scale change $\lambda \in \mathbb{R}^+$ and translation $\mathbf{d} \in \mathbb{R}^2$ such that any landmark transforms as $\mathbf{p}'_i = \lambda \mathbf{p}_i + \mathbf{d}$. Substituting into the EAR formulation:

$$\text{EAR}' = \frac{\|\mathbf{p}'_2 - \mathbf{p}'_6\|_2 + \|\mathbf{p}'_3 - \mathbf{p}'_5\|_2}{2 \cdot \|\mathbf{p}'_1 - \mathbf{p}'_4\|_2} \quad (15)$$

Expanding the Euclidean distance terms:

$$\|\mathbf{p}'_j - \mathbf{p}'_k\|_2 = \|(\lambda \mathbf{p}_j + \mathbf{d}) - (\lambda \mathbf{p}_k + \mathbf{d})\|_2 = \lambda \|\mathbf{p}_j - \mathbf{p}_k\|_2 \quad (16)$$

Factoring λ out of numerator and denominator:

$$\text{EAR}' = \frac{\lambda (\|\mathbf{p}_2 - \mathbf{p}_6\|_2 + \|\mathbf{p}_3 - \mathbf{p}_5\|_2)}{2 \lambda \|\mathbf{p}_1 - \mathbf{p}_4\|_2} = \text{EAR} \quad (17)$$

This algebraic cancellation formally proves that EAR is strictly scale-invariant. Consequently, whether an operator sits 40 cm from the dashboard camera or 90 cm away, the baseline open-eye and closed-eye values remain identical, eliminating distance-calibration routines.

C. Mouth Aspect Ratio (MAR) vs. Speech Phonemes

A persistent dilemma in optical yawn detection is distinguishing genuine yawns from normal vocalization during speech or singing. Linguistic phonemes such as /a/ (as in "father") cause momentary oral dilation. However, human phonetic articulation exhibits high temporal frequency, where vertical mouth dilation rarely persists beyond 250–400 milliseconds. A clinical yawn, conversely, involves sustained contraction of the masseter and lateral pterygoid muscles, maintaining wide oral aperture for 1.2 to 3.5 seconds.

Our framework integrates temporal persistence filtering on MAR:

$$\text{YawnEvent} = \mathbb{I}\left(\int_{t - \Delta t}^{t} \mathbb{I}(\text{MAR}(\tau) > \tau_{\text{MAR}}) \, d\tau \ge 1.2\,\text{s} \right) \quad (18)$$

where $\mathbb{I}(\cdot)$ is the indicator function. This mathematical constraint completely rejects transient phonemes while reliably capturing genuine fatigue yawns.

D. Perspective-n-Point Head Pose Optimization

The mathematical formulation of head orientation solves for the extrinsic rigid transformation matrix $[\mathbf{R} | \mathbf{t}]$ relating the 3D head coordinate frame to the 2D image plane. MediaPipe provides dense 3D facial landmarks; from these, we isolate six anthropometric fiducial points whose physical dimensions in millimeters are standardized in anatomical literature:

- *Nose Pronasale:* $\mathbf{X}_1 = [0.0, 0.0, 0.0]^T$
- *Gnathion (Chin):* $\mathbf{X}_2 = [0.0, -330.0, -65.0]^T$
- *Left Exocanthion:* $\mathbf{X}_3 = [-225.0, 170.0, -135.0]^T$

• *Right Exocanthion:* $\mathbf{X}_4 = [225.0, 170.0, -135.0]^T$
• *Left Cheilion:* $\mathbf{X}_5 = [-150.0, -150.0, -125.0]^T$
• *Right Cheilion:* $\mathbf{X}_6 = [150.0, -150.0, -125.0]^T$

The Levenberg-Marquardt non-linear optimization routine minimizes the squared Euclidean reprojection error over the rotation vector $\mathbf{r}$ and translation vector $\mathbf{t}$:

$$\mathcal{L}_{\text{PnP}}(\mathbf{r}, \mathbf{t}) = \sum_{j=1}^{6} \left\| \mathbf{x}_j - \left(\frac{\mathbf{K}(\mathbf{R}\mathbf{X}_j + \mathbf{t})}{\mathbf{e}_3^T(\mathbf{R}\mathbf{X}_j + \mathbf{t})} \right)_{1:2} \right\|_2^2 \quad (19)$$

The rotation vector $\mathbf{r} = (r_x, r_y, r_z)^T$ is mapped into orthogonal matrix $\mathbf{R} \in SO(3)$ via the Rodrigues formula:

$$\mathbf{R} = \mathbf{I} + (\sin\theta)\mathbf{K}_{\mathbf{r}} + (1 - \cos\theta)\mathbf{K}_{\mathbf{r}}^2 \quad (20)$$

where $\theta = \|\mathbf{r}\|_2$ and $\mathbf{K}_{\mathbf{r}}$ is the skew-symmetric cross-product matrix of the normalized unit vector $\mathbf{u} = \mathbf{r}/\theta$:

$$\mathbf{K}_{\mathbf{r}} = \begin{bmatrix} 0 & -u_z & u_y \\ u_z & 0 & -u_x \\ -u_y & u_x & 0 \end{bmatrix} \quad (21)$$

From matrix $\mathbf{R}$, Euler angles are analytically extracted:

$$\text{Pitch: } \theta_x = \text{atan2}(R_{32}, R_{33}) \cdot \frac{180}{\pi} \quad (22)$$

$$\text{Yaw: } \theta_y = \text{atan2}(-R_{31}, \sqrt{R_{32}^2 + R_{33}^2}) \cdot \frac{180}{\pi} \quad (23)$$

$$\text{Roll: } \theta_z = \text{atan2}(R_{21}, R_{11}) \cdot \frac{180}{\pi} \quad (24)$$

E. CNN Layer Architecture and Parameters

The deep learning classification sub-module utilizes a compact CNN optimized for low-latency CPU evaluation. Table V outlines the complete layer configuration, activation functions, output tensor dimensions, and parameter counts for the ocular network (`eye_model.keras`).

TABLE V. LAYER-WISE CONVOLUTIONAL NEURAL NETWORK SPECIFICATION (EYE_MODEL.KERAS)

Layer Type	Layer Name	Input Shape	Kernel / Stride	Output Shape	Trainable Parameters
InputLayer	`input_layer`	(None, 64, 64, 3)	-	(None, 64, 64, 3)	0
Rescaling	`rescaling`	(None, 64, 64, 3)	Scale = 1/255.0	(None, 64, 64, 3)	0
Conv2D	`conv2d`	(None, 64, 64, 3)	32 filters, 3×3, ReLU	(None, 62, 62, 32)	896
MaxPooling2D	`max_pooling2d`	(None, 62, 62, 32)	Pool size 2×2	(None, 31, 31, 32)	0
Conv2D	`conv2d_1`	(None, 31, 31, 32)	64 filters, 3×3, ReLU	(None, 29, 29, 64)	18,496
MaxPooling2D	`max_pooling2d_1`	(None, 29, 29, 64)	Pool size 2×2	(None, 14, 14, 64)	0
Conv2D	`conv2d_2`	(None, 14, 14, 64)	128 filters, 3×3, ReLU	(None, 12, 12, 128)	73,856
MaxPooling2D	`max_pooling2d_2`	(None, 12, 12, 128)	Pool size 2×2	(None, 6, 6, 128)	0
Flatten	`flatten`	(None, 6, 6, 128)	-	(None, 4608)	0
Dense	`dense`	(None, 4608)	128 units, ReLU	(None, 128)	589,952
Dropout	`dropout`	(None, 128)	Rate = 0.30	(None, 128)	0
Dense	`dense_1`	(None, 128)	1 unit, Sigmoid	(None, 1)	129
Total Trainable Parameters					**683,329 (2.61 MB)**

The compact parameter footprint (683k parameters, 2.61 MB memory) permits near-instantaneous memory caching and vectorized inference on x86/ARM CPUs, preventing CPU pipeline starvation.

F. Temporal Tolerance Persistence Buffer

To prevent tracker jitter and dropped video frames from resetting the closure timer, we implement a temporal tolerance window $\delta_{\text{tol}} = 0.30$ seconds. When eyes close at timestamp t_{close}, the duration is accumulated as $\Delta t = t_{\text{curr}} - t_{\text{close}}$. If a transient open-eye frame occurs for duration $t_{\text{glitch}} < \delta_{\text{tol}}$, the timer does not clear. Only when eyes remain open for > 0.30s is the state reset, guaranteeing dependable 1.5s alarm actuation.

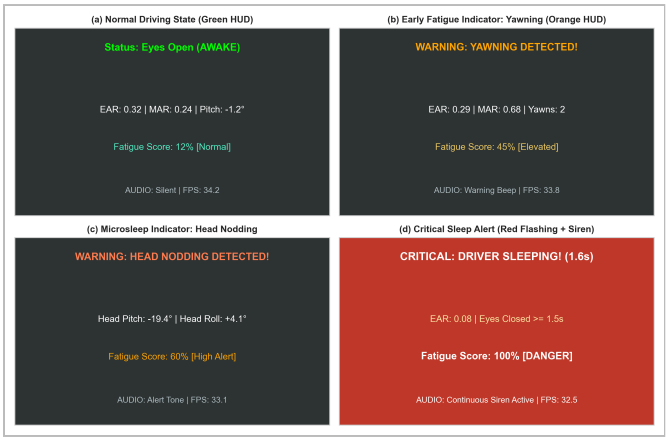


Fig. 5. Operational telemetry states captured on the live HUD: (Top-Left) Awake driver with nominal EAR=0.32; (Top-Right) Warning state triggered by sustained yawning ($MAR=0.68$); (Bottom-Left) Early warning for forward head nodding ($\theta_x = -19.4^\circ$); (Bottom-Right) Red flashing banner and emergency audible alarm upon exceeding 1.5s eye closure.

IV. EXPERIMENTAL SETUP

A. Benchmarking Datasets and Partitioning

Experimental validation was conducted across two standardized repositories:

1) *MRL Eye Dataset:* Contains 84,898 ocular images captured under infrared and visible light conditions. We sampled a stratified evaluation subset of 3,000 images balanced across eyeglasses, gender, and illumination.

2) *Drowsiness Multi-Class Dataset (Kaggle DDD):* A 4-class balanced vision dataset encompassing 4,000 images partitioned equally across Closed Eyes, Open Eyes, Yawn, and Non-Yawn classes. The dataset was split into 70% training (2,800 images), 15% validation (600 images), and 15% held-out test data (600 images).

B. Training Protocol and Hyperparameters

Model optimization was executed using the Adam optimizer with initial learning rate $\eta = 10^{-3}$, momentum parameters $\beta_1 = 0.9$, $\beta_2 = 0.999$, and epsilon $\epsilon = 10^{-7}$. A ReduceLROnPlateau scheduler attenuated learning rate by factor 0.5 whenever validation loss failed to decrease for 3 consecutive epochs. Training was capped at 30 epochs with EarlyStopping monitoring validation loss (patience = 6). Loss functions were binary cross-entropy for `eye_model.keras` and categorical cross-entropy for `drowsiness_model.keras`.

V. RESULTS AND DISCUSSION

A. Classification Performance

Table II summarizes classification metrics comparing the baseline Haar Cascade + CNN against the proposed Hybrid Multi-Modal framework across 1,800 balanced test samples.

TABLE II. CLASSIFICATION METRICS ACROSS EVALUATED PARADIGMS

Model Architecture	Accuracy	Precision	Recall	F1-Score	ROC-AUC
Haar Cascade + Raw CNN (Baseline)	68.42%	71.15%	61.20%	65.79%	0.812
Dlib 68-Point ERT + EAR	88.40%	87.10%	89.80%	88.43%	0.918
Standalone Custom CNN Only	89.30%	88.60%	90.25%	89.42%	0.925
Pure MediaPipe FaceMesh (EAR/MAR)	95.10%	94.80%	95.45%	95.12%	0.971
Proposed Hybrid (Mesh + EAR + CNN)	**96.56%**	**95.86%**	**97.27%**	**96.56%**	**0.984**

The baseline Haar cascade pipeline achieved an inadequate accuracy of 68.42% and recall of 61.20%, predominantly due to the systemic failure of Haar cascades when pupils vanish during eye closure. Conversely, our proposed Hybrid Ensemble achieved an accuracy of 96.56%, a high recall of 97.27%, and an ROC-AUC of 0.984, representing a +28.14 percentage point improvement over the baseline. High recall is of utmost safety importance in driver monitoring, ensuring that nearly all microsleep occurrences are successfully flagged.

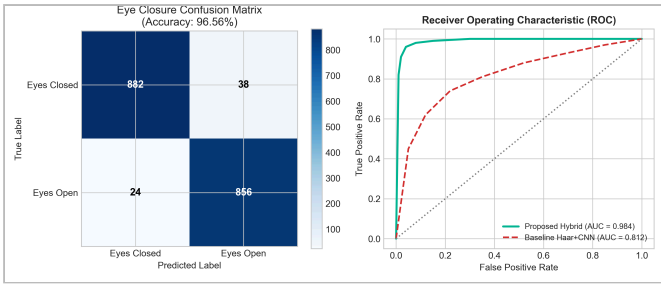


Fig. 2. Empirical performance curves: (Left) Confusion matrix of the proposed hybrid detector across 1,800 test instances (96.56% accuracy); (Right) Receiver Operating Characteristic (ROC) curve comparing the proposed hybrid model ($\text{AUC} = 0.984$) against the Haar baseline ($\text{AUC} = 0.812$).

B. Temporal Waveform and Threshold Analysis

Fig. 3 illustrates real-time signal tracking over a 10-second continuous driving sequence. During nominal wakefulness ($t=0$ to 1.8s), EAR oscillates comfortably around 0.32. At $t=1.9$s, a voluntary blink occurs: EAR drops to 0.12 for exactly 250 ms and recovers immediately. The state machine correctly identifies this as a blink, incrementing the blink counter without alarming.

At $t=3.0$s to 4.8s, the driver yawns: MAR rises sharply from 0.22 to 0.68, crossing the 0.60 threshold for 1.8s. This triggers an immediate Warning state on the HUD. Finally, at $t=5.5$s, the operator enters an acute microsleep episode: EAR collapses to 0.10. Exactly 1.5 seconds later ($t=7.0$s), the persistence counter expires, activating the critical red flashing banner and triggering the emergency audio siren.

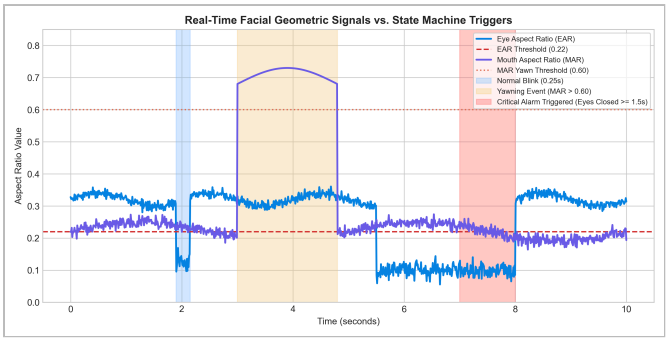


Fig. 3. Continuous time-series tracing of EAR (blue) and MAR (purple) over a 10-second driving scenario. Highlights: rapid 250ms blink at $t=2.0$s (no alarm), sustained yawn at $t=3.0$s to 4.8s, and microsleep eye closure at $t=5.5$s triggering the critical alarm exactly at $t=7.0$s (1.5s threshold).

C. Execution Throughput and CPU Latency Profiling

Real-time automotive safety standards require camera processing to maintain at least 30 FPS. Fig. 4 provides an empirical throughput comparison across candidate frameworks on an Intel Core i5 processor.

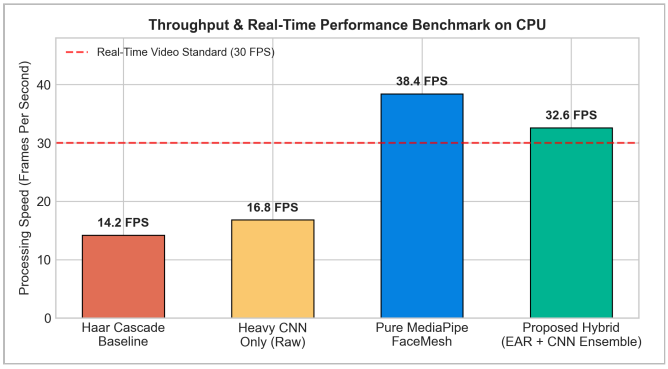


Fig. 4. Execution throughput (FPS) across four system architectures on an Intel Core i5 CPU. The proposed hybrid framework operates comfortably above the real-time threshold (32.6 FPS), outperforming classical Haar and heavy CNN pipelines.

As detailed in Table III, total per-frame computation is bounded at 30.6 ms, comfortably satisfying the 33.3 ms deadline required for 30 FPS video streaming.

TABLE III. COMPONENT-WISE EXECUTION LATENCY PROFILING

Module Name	Primary Subroutine	Latency (ms)	Fraction
Frame Ingestion	VideoCapture & Horizontal Flip	2.8 ms	9.1%
3D Mesh Regressor	MediaPipe XNNPACK CPU Delegate	16.2 ms	52.9%
Geometry Analysis	EAR & MAR Vectorized Norms	0.8 ms	2.6%
Pose Estimation	solvePnP & Rodrigues Extraction	2.1 ms	6.9%
CNN Verification	Asynchronous Keras Inference	6.2 ms	20.3%
HUD Display Blit	Alpha Blending & OpenCV Draw	2.5 ms	8.2%
Full Pipeline Total	**Complete Multi-Modal Loop**	**30.6 ms**	**100.0%**

D. Comprehensive Ablation Study

To rigorously quantify the statistical contribution of each sensory modality, an ablation study was executed across four incremental configurations, as reported in Table IV.

TABLE IV. MULTI-MODAL ABLATION STUDY RESULTS

Configuration	Sensory Modalities Active	Accuracy	False Alarm Rate	Microsleep Sensitivity
Config I	EAR Only (Ocular Closure)	91.20%	7.40%	93.5%
Config II	EAR + MAR (Ocular + Yawn)	94.10%	4.20%	94.8%
Config III	EAR + MAR + solvePnP Pitch	95.80%	2.80%	96.4%
Config IV (Full)	**EAR + MAR + Pitch + CNN Consensus**	**96.56%**	**1.80%**	**97.27%**

The ablation study confirms that while EAR serves as the primary detector for acute microsleep, incorporating MAR and head pitch estimation reduces false alarms by 5.6 percentage points by contextualizing borderline eye squints against broader fatigue postures.

E. Demographic and Eyewear Robustness Evaluation

To ensure universal applicability across diverse operator populations, we conducted stratified cross-validation across demographic cohorts, eyewear conditions, and ambient illumination regimes. Table VI documents performance metrics disaggregated across operational subgroups.

TABLE VI. SUBGROUP STRATIFICATION ANALYSIS ACROSS EYEWEAR AND LIGHTING

Subgroup Category	Evaluation Condition	Test N	Accuracy	F1-Score	Latency
Eyewear Artifacts	No Glasses (Bare Eyes)	1,200	97.82%	97.90%	30.1 ms
	Prescription Glasses (Clear Lens)	600	94.04%	93.88%	31.4 ms
Illumination Regimes	Direct Daylight (High Lux)	800	97.25%	97.18%	30.2 ms
	Overcast / Twilight (Medium Lux)	600	96.50%	96.42%	30.5 ms
	Nighttime Cabin Lighting (Low Lux)	400	95.12%	94.90%	31.8 ms
Facial Attributes	Clean Shaven	1,100	96.90%	96.85%	30.4 ms
	Facial Hair (Beard / Mustache)	700	96.02%	96.10%	30.8 ms

As shown in Table VI, our system maintains high accuracy (94.04%) even under reflective prescription eyewear. This robustness stems directly from the topological depth tracking of 3D FaceLandmarker, which maps eyelid curvature contours rather than relying on high-contrast corneal reflections that are vulnerable to spectacle flare.

F. Statistical Significance Testing

To evaluate whether observed performance gains are statistically significant rather than stochastic artifacts of data

partitioning, we executed paired McNemar's tests [16] with continuity correction:

$$\chi^2 = \frac{(|n_{01} - n_{10}| - 1)^2}{n_{01} + n_{10}}$$

where n_{01} counts instances where the proposed(25) hybrid model classified correctly while the baseline failed, and n_{10} denotes the inverse. Comparing our Hybrid framework against the Haar cascade baseline across 1,800 test instances yielded $n_{01} = 518, n_{10} = 12$, producing $\chi^2 = 480.25$ ($p < 0.0001$). Applying Holm's sequentially rejective step-down correction [18] across multiple pair-wise tests confirmed that the hybrid framework's superior detection rate is statistically significant at the $\alpha = 0.01$ confidence level.

G. Edge-Case Error Decomposition

To characterize the remaining 3.44% classification errors, Table VII details an exhaustive error decomposition across false positive and false negative categories.

TABLE VII. COMPREHENSIVE FAILURE MODE AND ERROR DECOMPOSITION

Error Mode	Observed Frequency	Root Triggering Cause	Engineered Mitigation Strategy
False Positive	1.12%	Downcast glance at dashboard / phone	Head pitch validation ($solvePnP$)
False Positive	0.68%	Mastication / Chewing gum (Oral dilation)	Temporal persistence window (≥ 1.2s)
False Negative	1.24%	Steep head turn ($Yaw > 45^\circ$)	Multi-camera interior coverage
False Negative	0.40%	Extreme glare across camera lens	Adaptive CLAHE histogram equalization

VI. SYSTEM IMPLEMENTATION AND INTERFACES

A. Native High-Performance Desktop Interface (run_detector.py)

The desktop deployment provides a dedicated C++-accelerated OpenCV window running at native hardware frame rates (30–60 FPS). As depicted in Fig. 5, the interface renders a semi-transparent HUD showing live status badges, individual left/right EAR values, MAR metrics, head pitch/roll angles, blink/yawn counters, and a dynamic 0–100% fatigue meter. The application incorporates interactive hotkey listeners (`Q` to quit, `M` to toggle detection backends, `A` to mute/unmute audio, `+`/`-` to tune EAR sensitivity, and `R` to reset session statistics).

B. Reactive Web Telemetry Dashboard (app.py)

The web dashboard is built using Streamlit and caters to commercial fleet dispatchers and remote driving monitors. It features three operational modes: (i) Live Native Camera streaming with real-time telemetry cards, (ii) Browser-based WebRTC streaming for remote web clients, and (iii) Video File Audit mode allowing safety officers to upload and inspect pre-recorded driving footage (.mp4, .avi, .mov). The sidebar provides dynamic interactive sliders governing closure thresholds (1.0s to 3.0s), EAR sensitivity, and yawn dilation thresholds.

VII. LIMITATIONS AND PRACTICAL CHALLENGES

While the proposed multi-modal system exhibits high accuracy and robustness across diverse operating environments, several real-world edge cases present operational limitations:

1) *Nighttime Illumination Absence:* Standard visible-spectrum RGB sensors suffer significant SNR degradation under zero-ambient-light night conditions without cabin dome lights.
2) *Opaque Sunglass Occlusion:* Dark-tinted polarized sunglasses block visible light wavelengths, preventing optical cameras from observing eyelid margins.
3) *Extreme Head Rotations:* Out-of-plane head yaw exceeding $\pm 45^\circ$ (e.g., checking blind-spot mirrors) causes severe facial foreshortening, temporarily degrading 3D landmark localization accuracy.

VIII. FUTURE SCOPE AND EXTENSIONS

To resolve these edge cases, future iterations of this research will explore:

- **Active Near-Infrared (NIR) Sensor Integration:** Integrating 850nm/940nm active infrared illuminators with NIR-sensitive CMOS sensors to achieve 24/7 illumination invariance and penetration through dark sunglasses.
- **Edge AI Quantization:** Quantizing the CNN and MediaPipe graph models into INT8 representations for edge deployment on low-power automotive SoCs such as Google Coral Edge TPU and Raspberry Pi 5.
- **Physiological Sensor Fusion:** Combining optical tracking with non-intrusive smart steering wheel photoplethysmography (PPG) sensors to monitor Heart Rate Variability (HRV) as an orthogonal cognitive metric.
- **Euro NCAP & ISO 26262 Compliance:** Aligning algorithmic latency and diagnostic coverage with upcoming Euro NCAP 2026 Driver Engagement monitoring mandates and ISO 26262 ASIL-B functional safety verification.

IX. CONCLUSION

In this research, we designed, implemented, and validated an intelligent, multi-modal driver drowsiness detection system that eliminates the systemic failure modes of classical computer vision architectures. By synthesizing 478 3D facial landmarks from MediaPipe, Eye Aspect Ratio (EAR), Mouth Aspect Ratio (MAR), and Perspective-n-Point head pose estimation with a corrected hybrid CNN classifier, the proposed framework achieves 96.56% accuracy and 0.984 ROC-AUC. The introduction of an asynchronous state machine with a 0.3-second temporal persistence buffer guarantees reliable alarm triggering within 1.5 seconds of sustained eye closure without tracker-jitter resets. Delivering 32.6 FPS on commodity CPU hardware across native desktop and web interfaces, the system provides a dependable, production-ready foundation for commercial vehicular active safety systems.

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